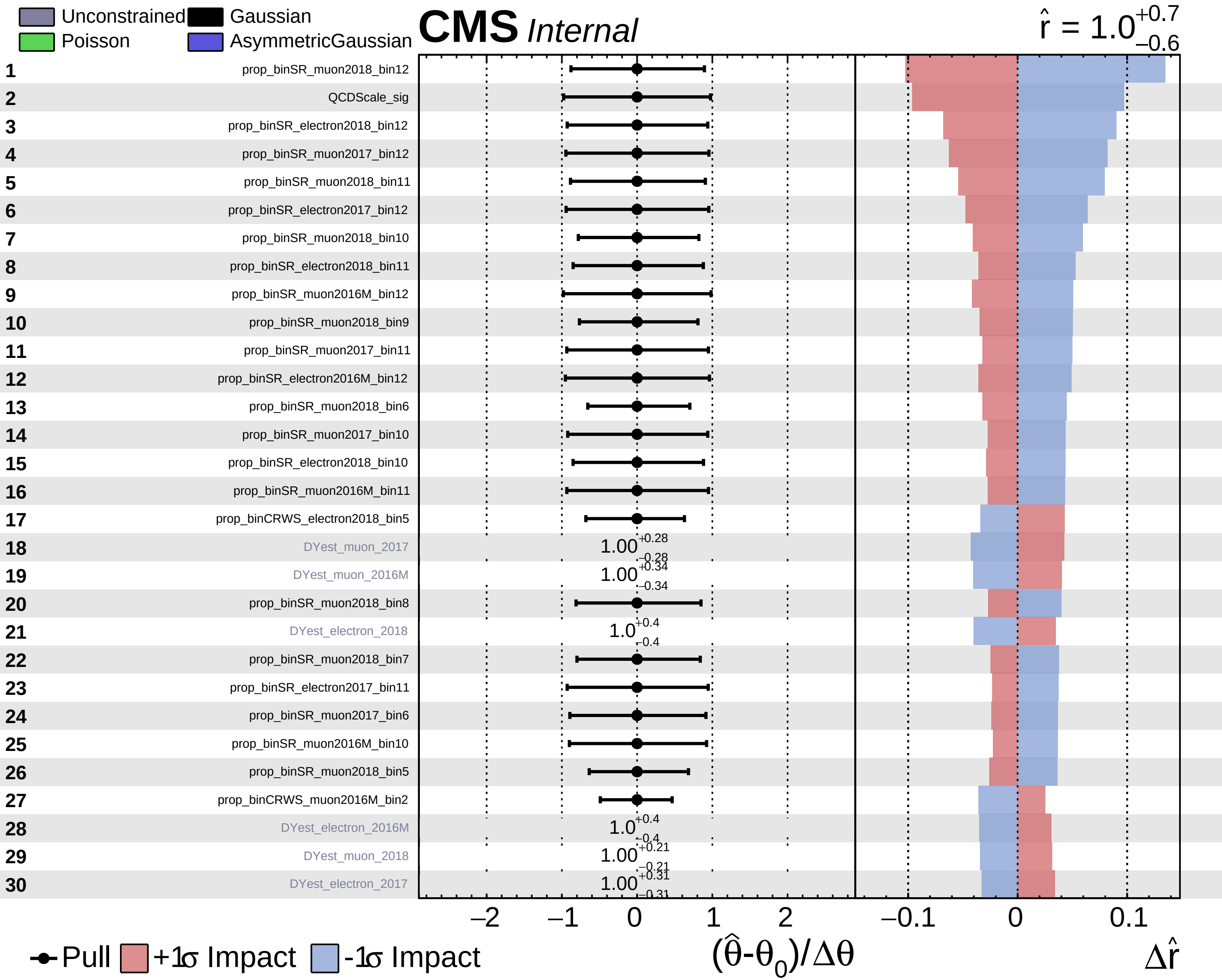
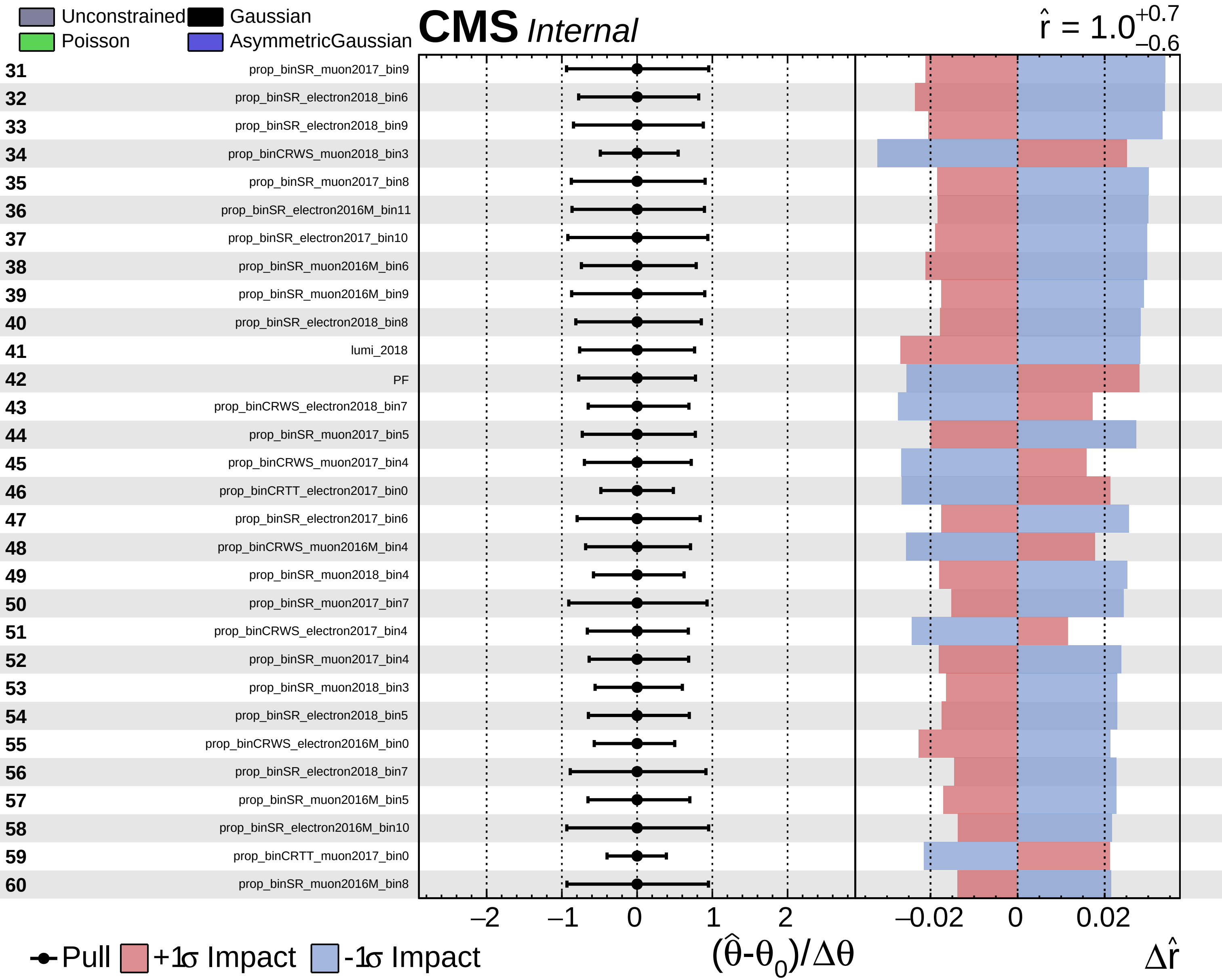
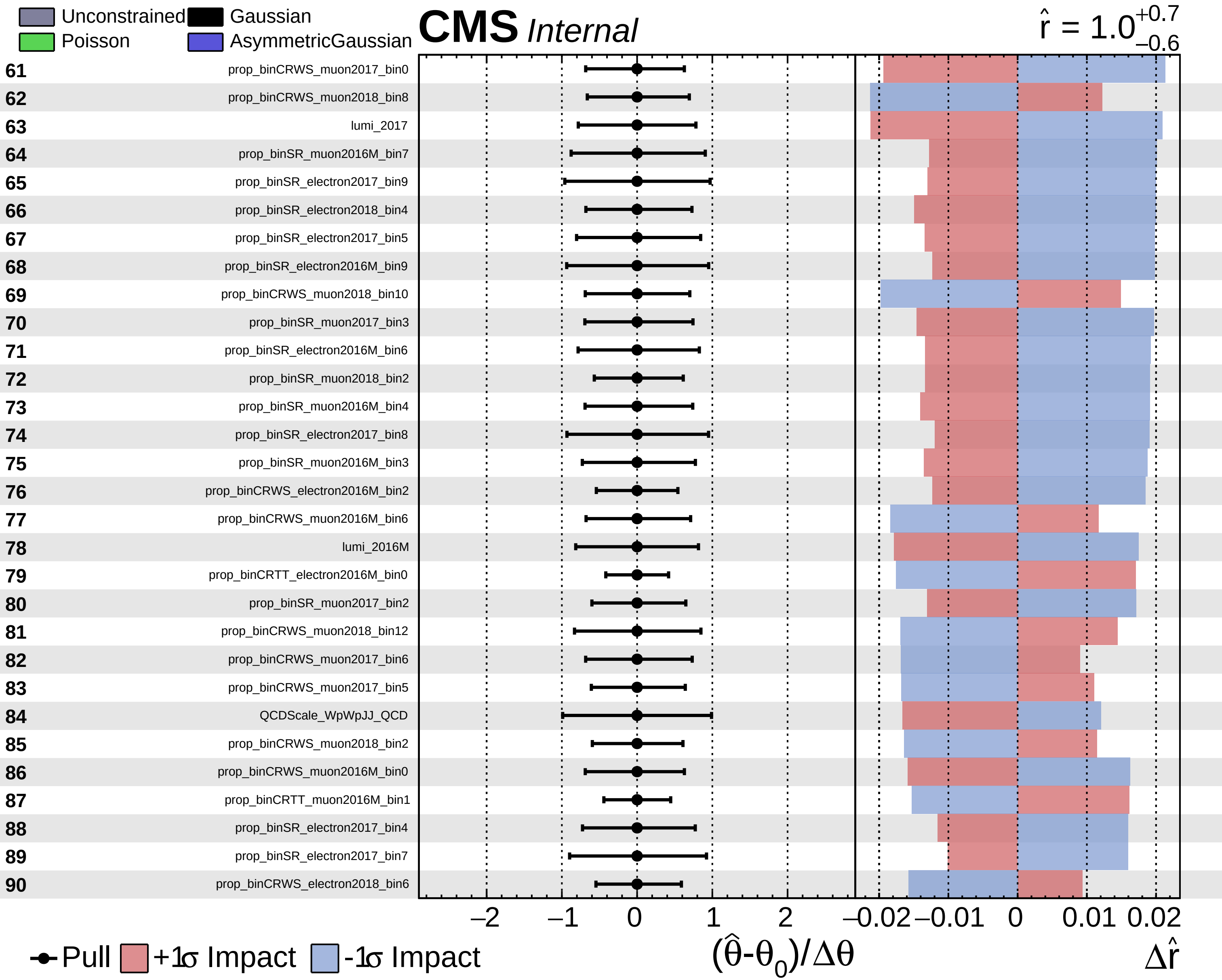


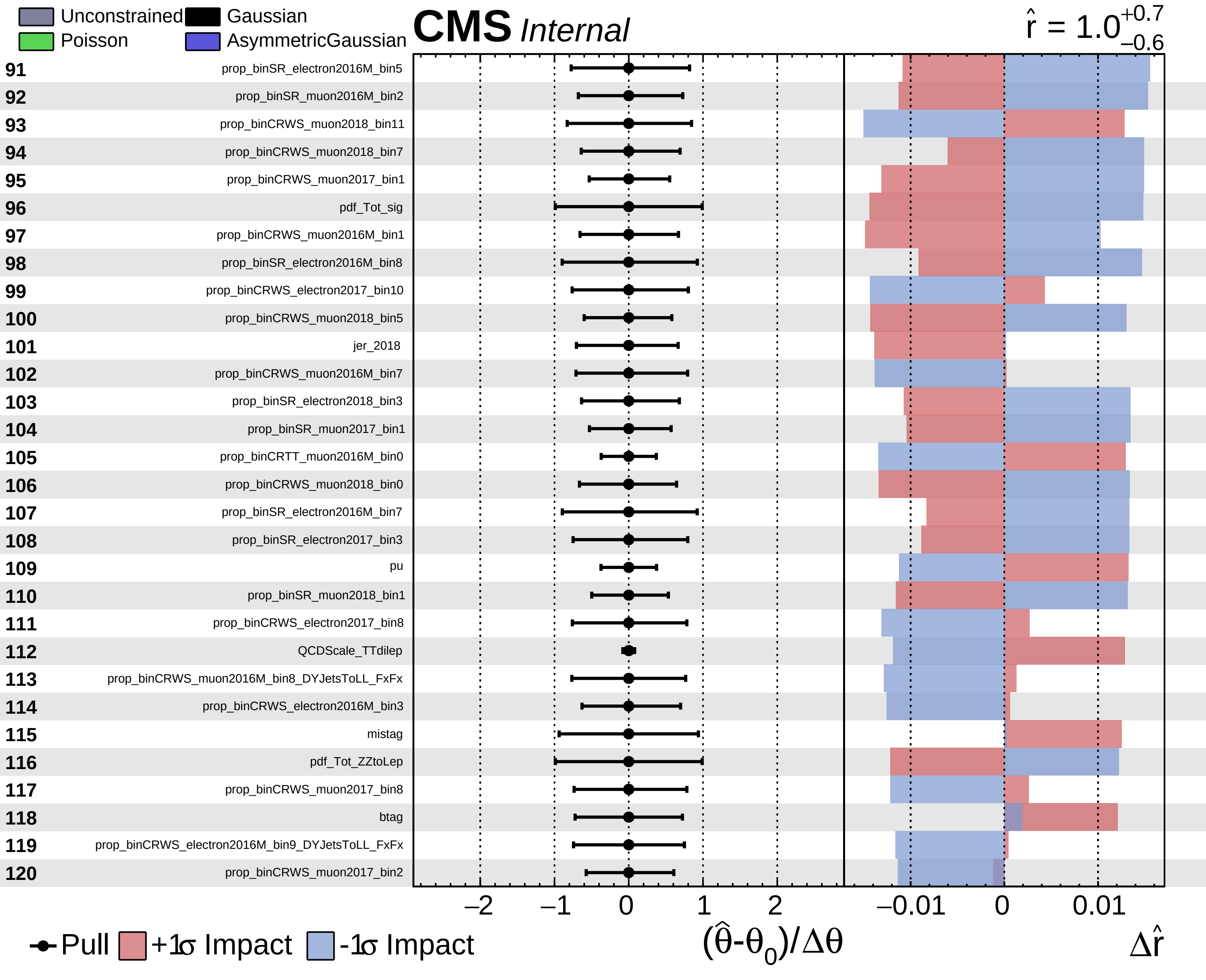
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.7}_{-0.6}$
1 prop_binSR_muon2018_bin12
2 QCDScale_sig
3 prop_binSR_electron2018_bin12
4 prop_binSR_muon2017_bin12
5 prop_binSR_muon2018_bin11
6 prop_binSR_electron2017_bin12
7 prop_binSR_muon2018_bin10
8 prop_binSR_electron2018_bin11
9 prop_binSR_muon2016M_bin12
10 prop_binSR_muon2018_bin9
11 prop_binSR_muon2017_bin11
12 prop_binSR_electron2016M_bin12
13 prop_binSR_muon2018_bin6
14 prop_binSR_muon2017_bin10
15 prop_binSR_electron2018_bin10
16 prop_binSR_muon2016M_bin11
17 prop_binCRWS_electron2018_bin5
18 DYest_muon_2017 $1.00^{+0.28}_{-0.28}$
19 DYest_muon_2016M $1.00^{+0.34}_{-0.34}$
20 prop_binSR_muon2018_bin8
21 DYest_electron_2018 $1.0^{+0.4}_{-0.4}$
22 prop_binSR_muon2018_bin7
23 prop_binSR_electron2017_bin11
24 prop_binSR_muon2017_bin6
25 prop_binSR_muon2016M_bin10
26 prop_binSR_muon2018_bin5
27 prop_binCRWS_muon2016M_bin2
28 DYest_electron_2016M $1.0^{+0.4}_{-0.4}$
29 DYest_muon_2018 $1.00^{+0.21}_{-0.21}$
30 DYest_electron_2017 $1.00^{+0.31}_{-0.31}$
−2 −1 0 1 2
−0.1 0 0.1
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



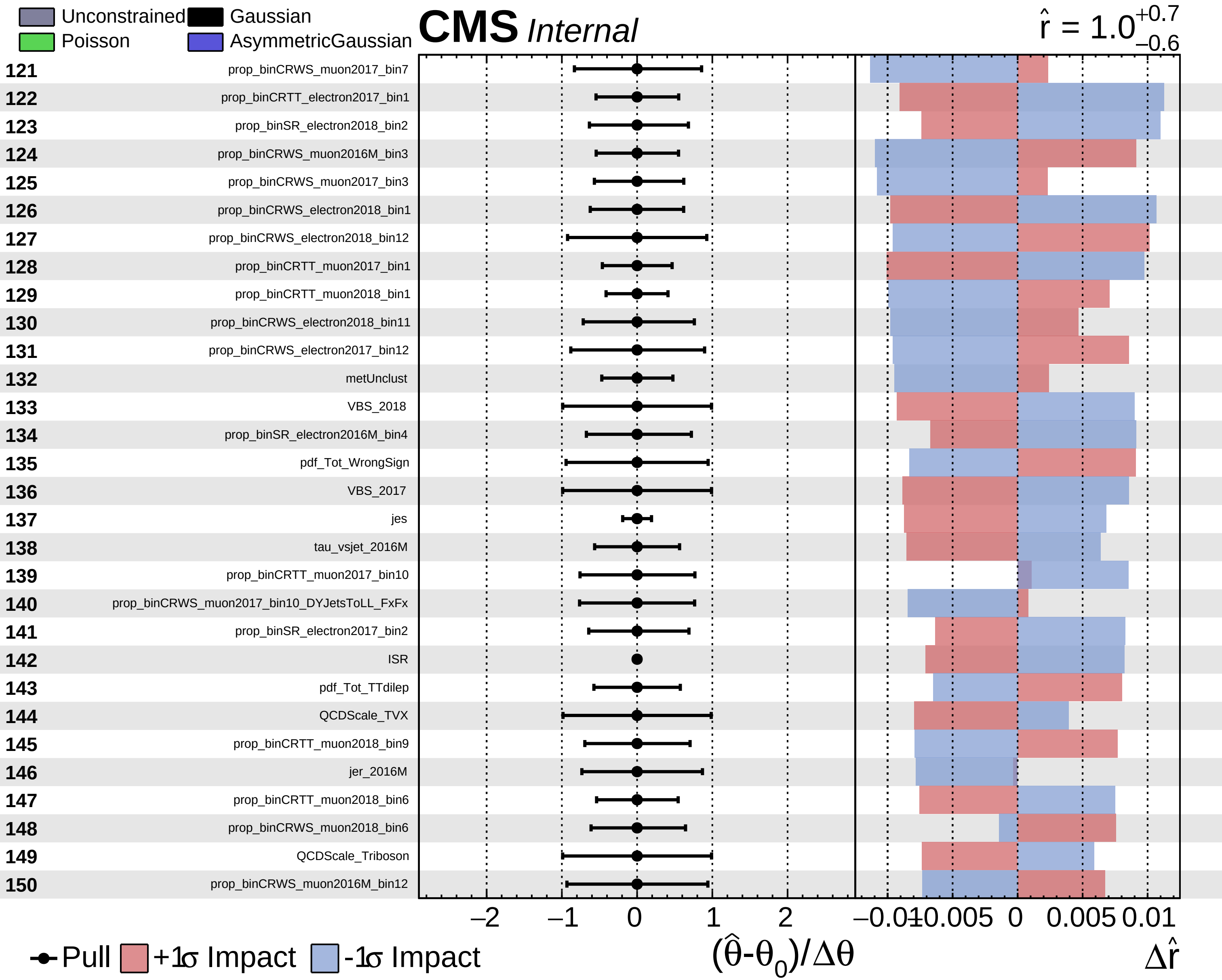
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.7}_{-0.6}$
31 prop_binSR_muon2017_bin9
32 prop_binSR_electron2018_bin6
33 prop_binSR_electron2018_bin9
34 prop_binCRWS_muon2018_bin3
35 prop_binSR_muon2017_bin8
36 prop_binSR_electron2016M_bin11
37 prop_binSR_electron2017_bin10
38 prop_binSR_muon2016M_bin6
39 prop_binSR_muon2016M_bin9
40 prop_binSR_electron2018_bin8
41 lumi_2018
42 PF
43 prop_binCRWS_electron2018_bin7
44 prop_binSR_muon2017_bin5
45 prop_binCRWS_muon2017_bin4
46 prop_binCRTT_electron2017_bin0
47 prop_binSR_electron2017_bin6
48 prop_binCRWS_muon2016M_bin4
49 prop_binSR_muon2018_bin4
50 prop_binSR_muon2017_bin7
51 prop_binCRWS_electron2017_bin4
52 prop_binSR_muon2017_bin4
53 prop_binSR_muon2018_bin3
54 prop_binSR_electron2018_bin5
55 prop_binCRWS_electron2016M_bin0
56 prop_binSR_electron2018_bin7
57 prop_binSR_muon2016M_bin5
58 prop_binSR_electron2016M_bin10
59 prop_binCRTT_muon2017_bin0
60 prop_binSR_muon2016M_bin8
−2
−1
0
1
2
−0.02
0
0.02
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

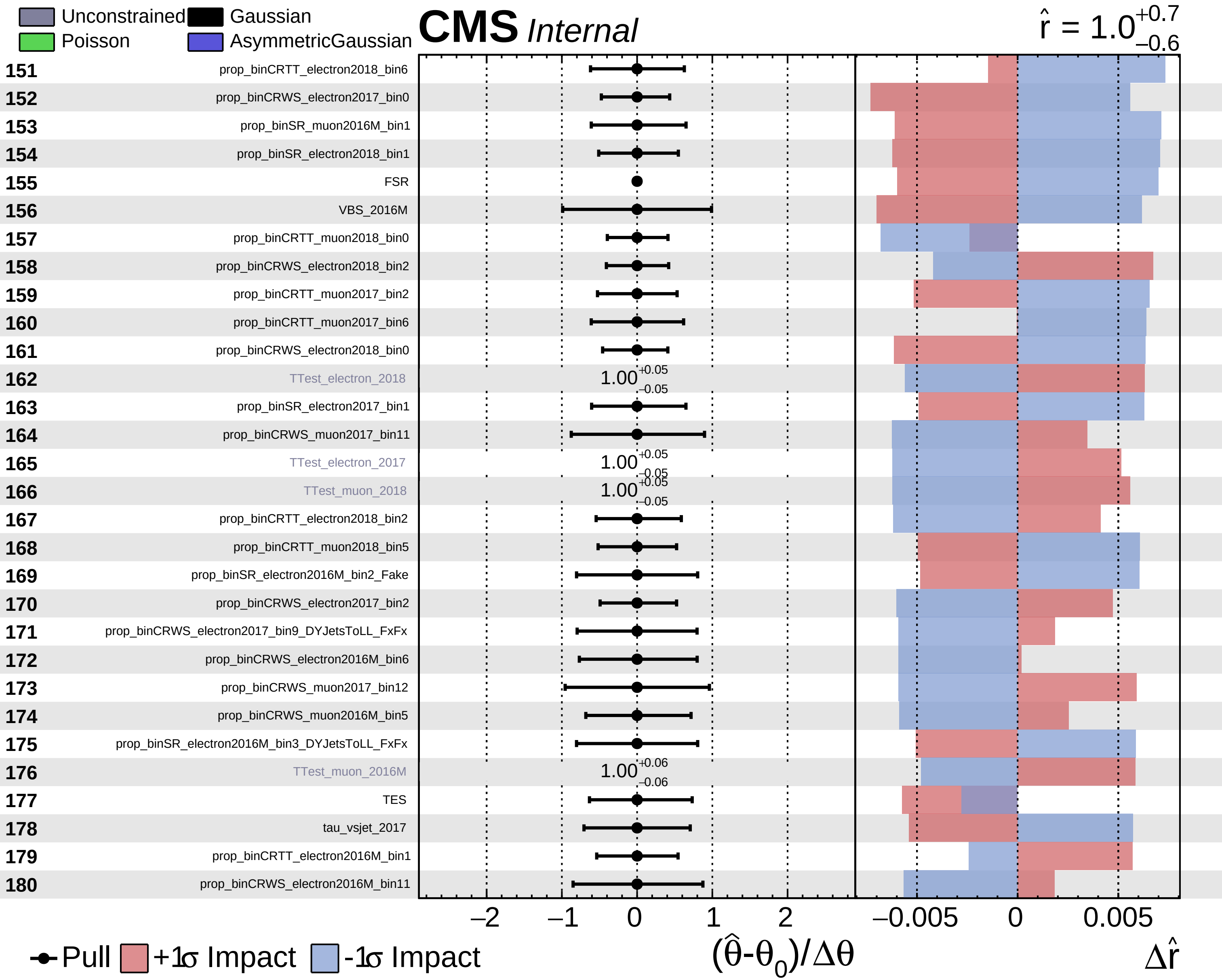


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.7}_{-0.6}$
61 prop_binCRWS_muon2017_bin0
62 prop_binCRWS_muon2018_bin8
63 lumi_2017
64 prop_binSR_muon2016M_bin7
65 prop_binSR_electron2017_bin9
66 prop_binSR_electron2018_bin4
67 prop_binSR_electron2017_bin5
68 prop_binSR_electron2016M_bin9
69 prop_binCRWS_muon2018_bin10
70 prop_binSR_muon2017_bin3
71 prop_binSR_electron2016M_bin6
72 prop_binSR_muon2018_bin2
73 prop_binSR_muon2016M_bin4
74 prop_binSR_electron2017_bin8
75 prop_binSR_muon2016M_bin3
76 prop_binCRWS_electron2016M_bin2
77 prop_binCRWS_muon2016M_bin6
78 lumi_2016M
79 prop_binCRTT_electron2016M_bin0
80 prop_binSR_muon2017_bin2
81 prop_binCRWS_muon2018_bin12
82 prop_binCRWS_muon2017_bin6
83 prop_binCRWS_muon2017_bin5
84 QCDScale_WpWpJJ_QCD
85 prop_binCRWS_muon2018_bin2
86 prop_binCRWS_muon2016M_bin0
87 prop_binCRTT_muon2016M_bin1
88 prop_binSR_electron2017_bin4
89 prop_binSR_electron2017_bin7
90 prop_binCRWS_electron2018_bin6
−2 −1 0 1 2
−0.02 −0.01 0 0.01 0.02
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

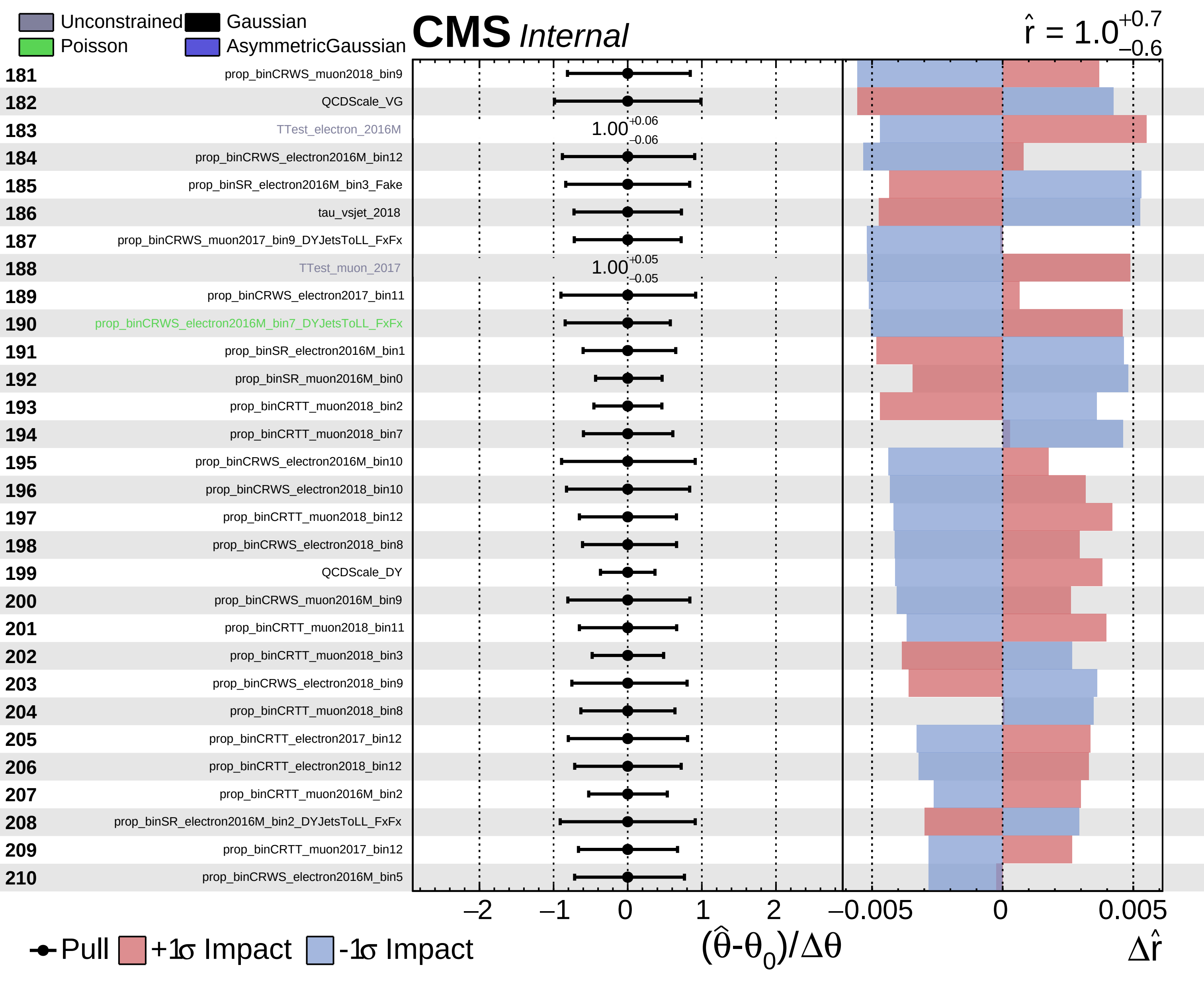


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.7}_{-0.6}$
91 prop_binSR_electron2016M_bin5
92 prop_binSR_muon2016M_bin2
93 prop_binCRWS_muon2018_bin11
94 prop_binCRWS_muon2018_bin7
95 prop_binCRWS_muon2017_bin1
96 pdf_Tot_sig
97 prop_binCRWS_muon2016M_bin1
98 prop_binSR_electron2016M_bin8
99 prop_binCRWS_electron2017_bin10
100 prop_binCRWS_muon2018_bin5
101 jer_2018
102 prop_binCRWS_muon2016M_bin7
103 prop_binSR_electron2018_bin3
104 prop_binSR_muon2017_bin1
105 prop_binCRTT_muon2016M_bin0
106 prop_binCRWS_muon2018_bin0
107 prop_binSR_electron2016M_bin7
108 prop_binSR_electron2017_bin3
109 pu
110 prop_binSR_muon2018_bin1
111 prop_binCRWS_electron2017_bin8
112 QCDScale_TTdilep
113 prop_binCRWS_muon2016M_bin8_DYJetsToLL_FxFx
114 prop_binCRWS_electron2016M_bin3
115 mistag
116 pdf_Tot_ZZtoLep
117 prop_binCRWS_muon2017_bin8
118 btag
119 prop_binCRWS_electron2016M_bin9_DYJetsToLL_FxFx
120 prop_binCRWS_muon2017_bin2
−2 −1 0 1 2
−0.01 0 0.01
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

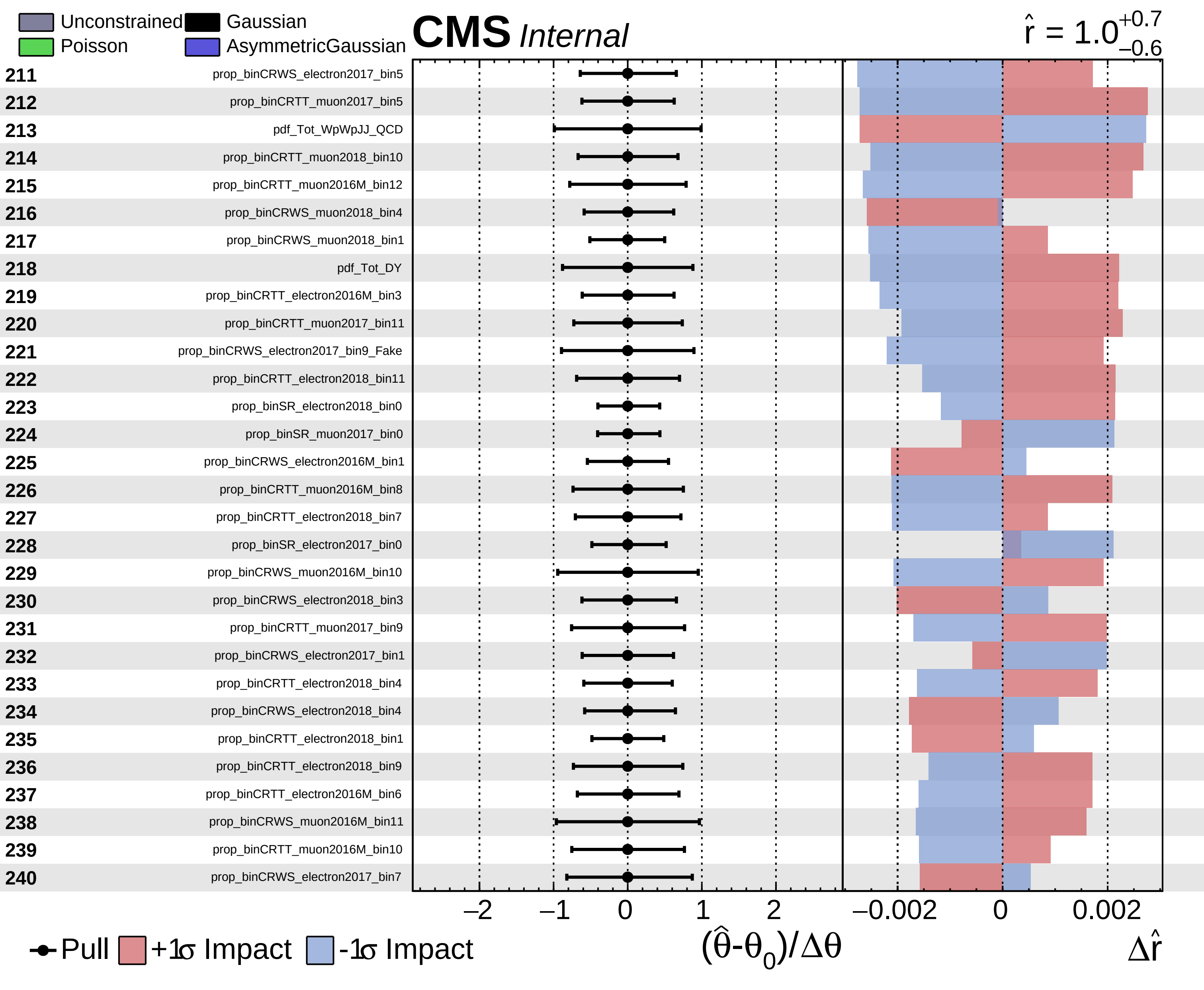

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.7}_{-0.6}$
121 prop_binCRWS_muon2017_bin7
122 prop_binCRTT_electron2017_bin1
123 prop_binSR_electron2018_bin2
124 prop_binCRWS_muon2016M_bin3
125 prop_binCRWS_muon2017_bin3
126 prop_binCRWS_electron2018_bin1
127 prop_binCRWS_electron2018_bin12
128 prop_binCRTT_muon2017_bin1
129 prop_binCRTT_muon2018_bin1
130 prop_binCRWS_electron2018_bin11
131 prop_binCRWS_electron2017_bin12
132 metUnclust
133 VBS_2018
134 prop_binSR_electron2016M_bin4
135 pdf_Tot_WrongSign
136 VBS_2017
137 jes
138 tau_vsjet_2016M
139 prop_binCRTT_muon2017_bin10
140 prop_binCRWS_muon2017_bin10_DYJetsToLL_FxFx
141 prop_binSR_electron2017_bin2
142 ISR
143 pdf_Tot_TTdilep
144 QCDScale_TVX
145 prop_binCRTT_muon2018_bin9
146 jer_2016M
147 prop_binCRTT_muon2018_bin6
148 prop_binCRWS_muon2018_bin6
149 QCDScale_Triboson
150 prop_binCRWS_muon2016M_bin12
−2 −1 0 1 2
−0.01 −0.005 0 0.005 0.01
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



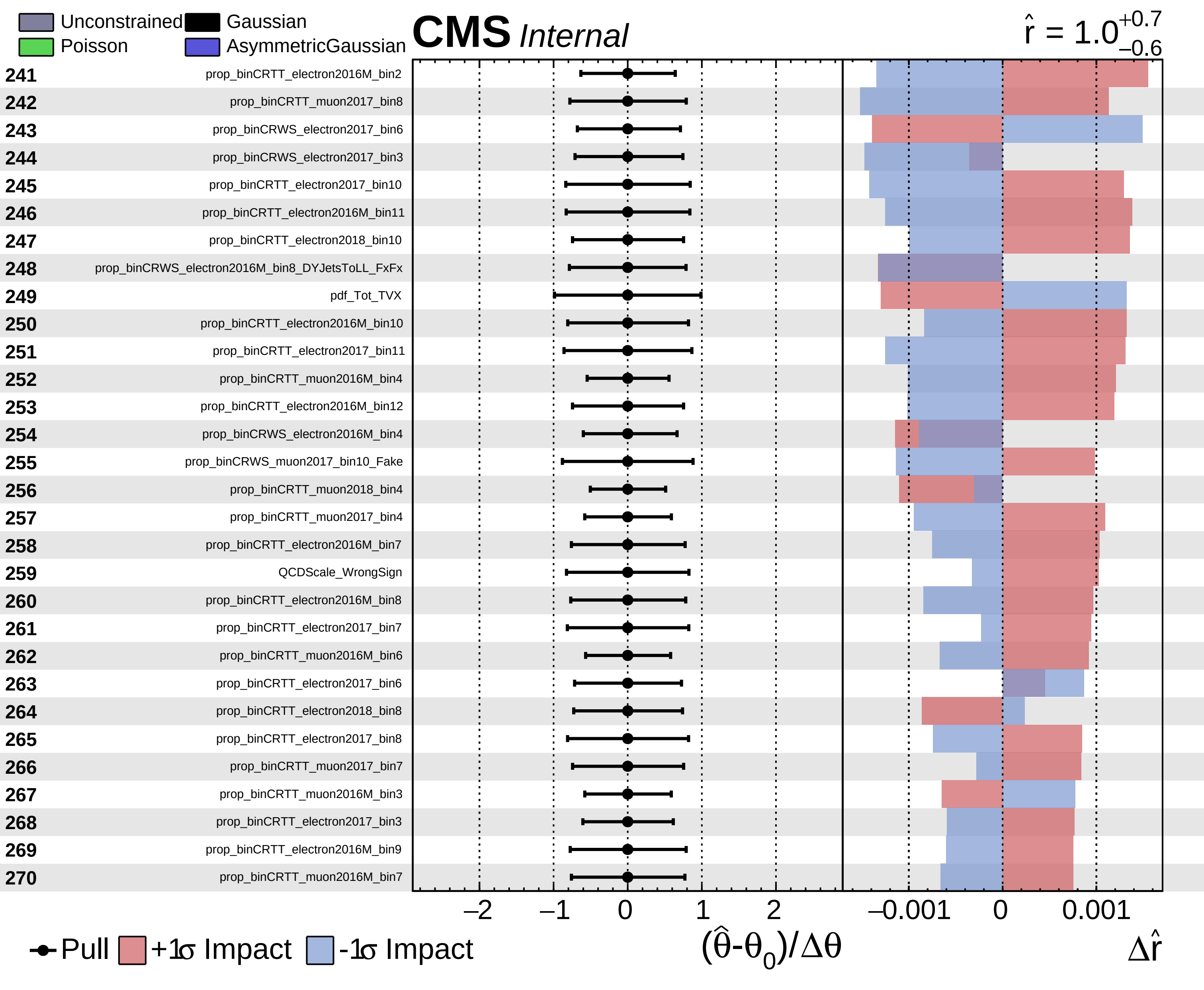
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.7}_{-0.6}$
151 prop_binCRTT_electron2018_bin6
152 prop_binCRWS_electron2017_bin0
153 prop_binSR_muon2016M_bin1
154 prop_binSR_electron2018_bin1
155 FSR
156 VBS_2016M
157 prop_binCRTT_muon2018_bin0
158 prop_binCRWS_electron2018_bin2
159 prop_binCRTT_muon2017_bin2
160 prop_binCRTT_muon2017_bin6
161 prop_binCRWS_electron2018_bin0
162 TTest_electron_2018 $1.00^{+0.05}_{-0.05}$
163 prop_binSR_electron2017_bin1
164 prop_binCRWS_muon2017_bin11
165 TTest_electron_2017 $1.00^{+0.05}_{-0.05}$
166 TTest_muon_2018 $1.00^{+0.05}_{-0.05}$
167 prop_binCRTT_electron2018_bin2
168 prop_binCRTT_muon2018_bin5
169 prop_binSR_electron2016M_bin2_Fake
170 prop_binCRWS_electron2017_bin2
171 prop_binCRWS_electron2017_bin9_DYJetsToLL_FxFx
172 prop_binCRWS_electron2016M_bin6
173 prop_binCRWS_muon2017_bin12
174 prop_binCRWS_muon2016M_bin5
175 prop_binSR_electron2016M_bin3_DYJetsToLL_FxFx
176 TTest_muon_2016M $1.00^{+0.06}_{-0.06}$
177 TES
178 tau_vsjet_2017
179 prop_binCRTT_electron2016M_bin1
180 prop_binCRWS_electron2016M_bin11
−2 −1 0 1 2
−0.005 0 0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

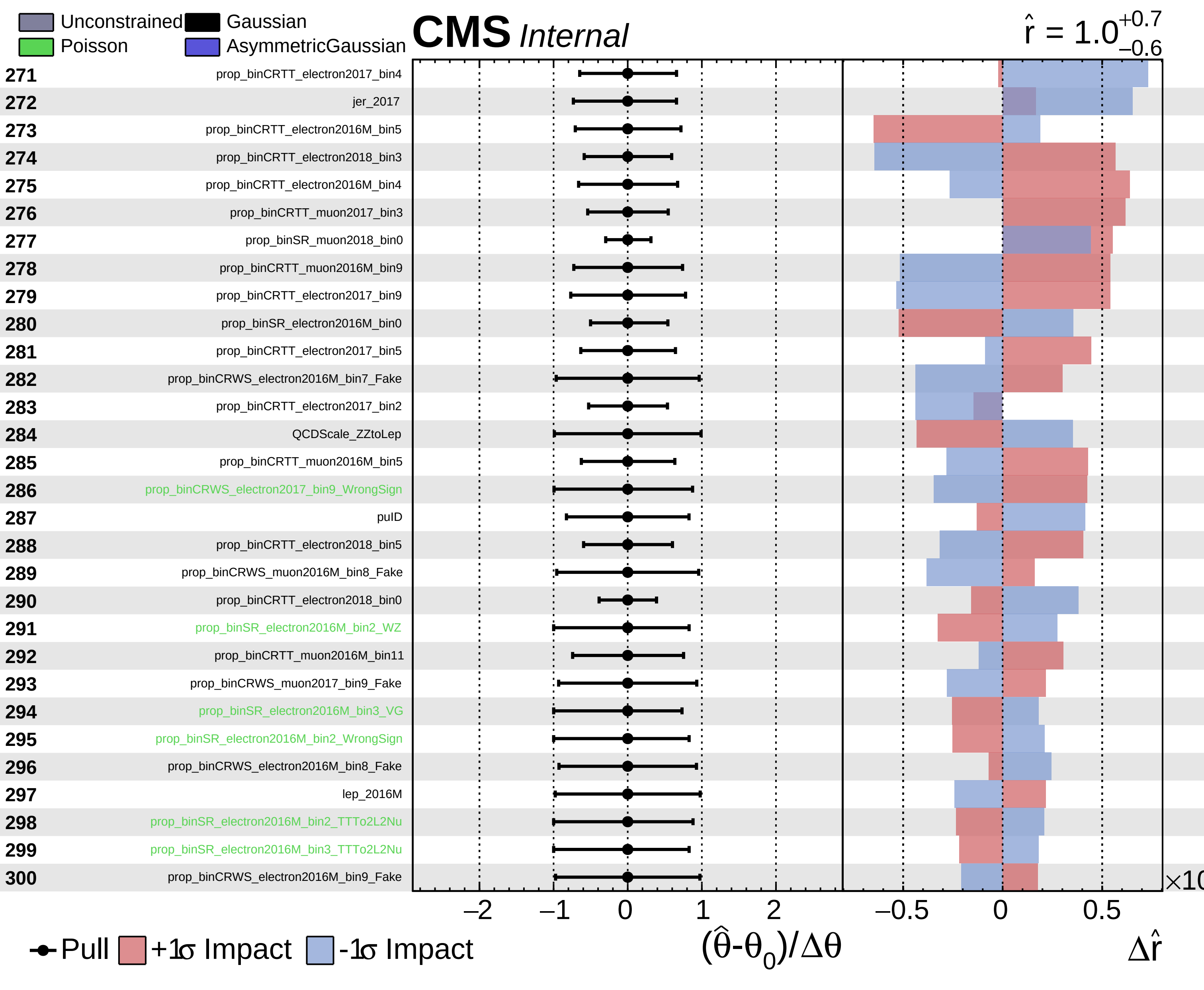


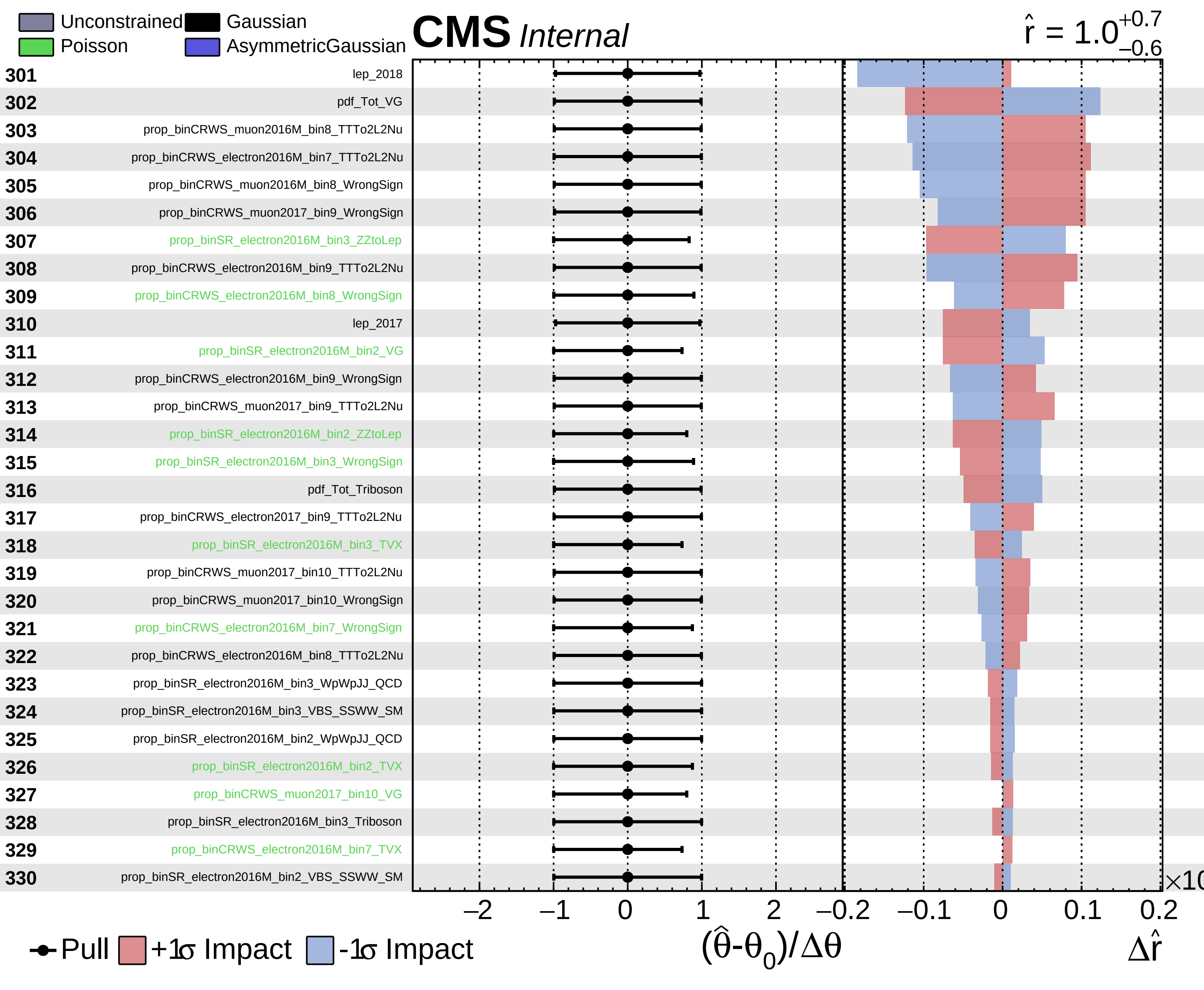
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 1.0 +0.7 −0.6
181 prop_binCRWS_muon2018_bin9
182 QCDScale_VG
183 TTest_electron_2016M
1.00 +0.06 −0.06
184 prop_binCRWS_electron2016M_bin12
185 prop_binSR_electron2016M_bin3_Fake
186 tau_vsjet_2018
187 prop_binCRWS_muon2017_bin9_DYJetsToLL_FxFx
188 TTest_muon_2017
1.00 +0.05 −0.05
189 prop_binCRWS_electron2017_bin11
190 prop_binCRWS_electron2016M_bin7_DYJetsToLL_FxFx
191 prop_binSR_electron2016M_bin1
192 prop_binSR_muon2016M_bin0
193 prop_binCRTT_muon2018_bin2
194 prop_binCRTT_muon2018_bin7
195 prop_binCRWS_electron2016M_bin10
196 prop_binCRWS_electron2018_bin10
197 prop_binCRTT_muon2018_bin12
198 prop_binCRWS_electron2018_bin8
199 QCDScale_DY
200 prop_binCRWS_muon2016M_bin9
201 prop_binCRTT_muon2018_bin11
202 prop_binCRTT_muon2018_bin3
203 prop_binCRWS_electron2018_bin9
204 prop_binCRTT_muon2018_bin8
205 prop_binCRTT_electron2017_bin12
206 prop_binCRTT_electron2018_bin12
207 prop_binCRTT_muon2016M_bin2
208 prop_binSR_electron2016M_bin2_DYJetsToLL_FxFx
209 prop_binCRTT_muon2017_bin12
210 prop_binCRWS_electron2016M_bin5
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂



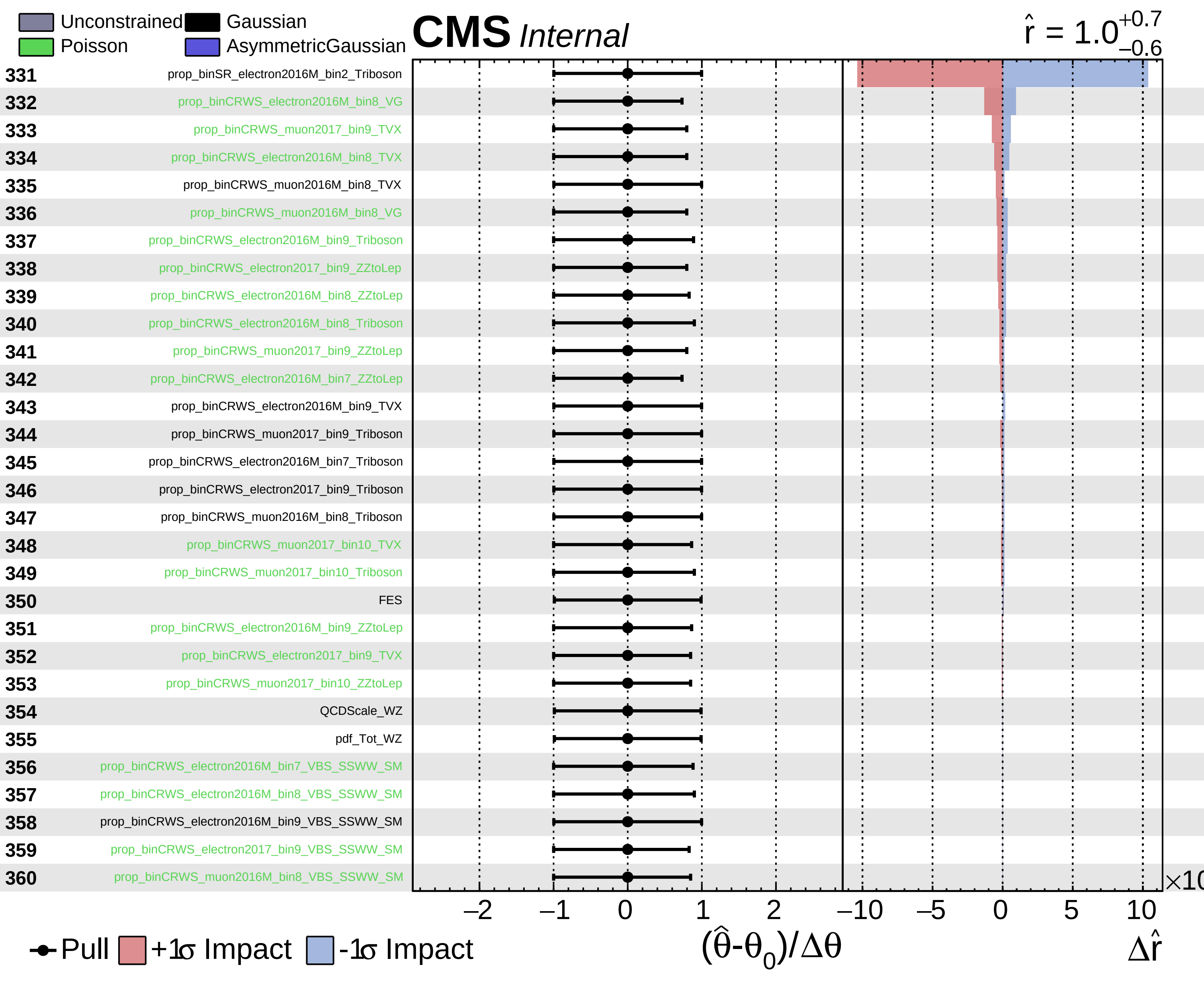
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.7}_{-0.6}$
211 prop_binCRWS_electron2017_bin5
212 prop_binCRTT_muon2017_bin5
213 pdf_Tot_WpWpJJ_QCD
214 prop_binCRTT_muon2018_bin10
215 prop_binCRTT_muon2016M_bin12
216 prop_binCRWS_muon2018_bin4
217 prop_binCRWS_muon2018_bin1
218 pdf_Tot_DY
219 prop_binCRTT_electron2016M_bin3
220 prop_binCRTT_muon2017_bin11
221 prop_binCRWS_electron2017_bin9_Fake
222 prop_binCRTT_electron2018_bin11
223 prop_binSR_electron2018_bin0
224 prop_binSR_muon2017_bin0
225 prop_binCRWS_electron2016M_bin1
226 prop_binCRTT_muon2016M_bin8
227 prop_binCRTT_electron2018_bin7
228 prop_binSR_electron2017_bin0
229 prop_binCRWS_muon2016M_bin10
230 prop_binCRWS_electron2018_bin3
231 prop_binCRTT_muon2017_bin9
232 prop_binCRWS_electron2017_bin1
233 prop_binCRTT_electron2018_bin4
234 prop_binCRWS_electron2018_bin4
235 prop_binCRTT_electron2018_bin1
236 prop_binCRTT_electron2018_bin9
237 prop_binCRTT_electron2016M_bin6
238 prop_binCRWS_muon2016M_bin11
239 prop_binCRTT_muon2016M_bin10
240 prop_binCRWS_electron2017_bin7
−2
−1
0
1
2
−0.002
0
0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.7}_{-0.6}$
241 prop_binCRTT_electron2016M_bin2
242 prop_binCRTT_muon2017_bin8
243 prop_binCRWS_electron2017_bin6
244 prop_binCRWS_electron2017_bin3
245 prop_binCRTT_electron2017_bin10
246 prop_binCRTT_electron2016M_bin11
247 prop_binCRTT_electron2018_bin10
248 prop_binCRWS_electron2016M_bin8_DYJetsToLL_FxFx
249 pdf_Tot_TVX
250 prop_binCRTT_electron2016M_bin10
251 prop_binCRTT_electron2017_bin11
252 prop_binCRTT_muon2016M_bin4
253 prop_binCRTT_electron2016M_bin12
254 prop_binCRWS_electron2016M_bin4
255 prop_binCRWS_muon2017_bin10_Fake
256 prop_binCRTT_muon2018_bin4
257 prop_binCRTT_muon2017_bin4
258 prop_binCRTT_electron2016M_bin7
259 QCDScale_WrongSign
260 prop_binCRTT_electron2016M_bin8
261 prop_binCRTT_electron2017_bin7
262 prop_binCRTT_muon2016M_bin6
263 prop_binCRTT_electron2017_bin6
264 prop_binCRTT_electron2018_bin8
265 prop_binCRTT_electron2017_bin8
266 prop_binCRTT_muon2017_bin7
267 prop_binCRTT_muon2016M_bin3
268 prop_binCRTT_electron2017_bin3
269 prop_binCRTT_electron2016M_bin9
270 prop_binCRTT_muon2016M_bin7
−2
−1
0
1
2
−0.001
0
0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

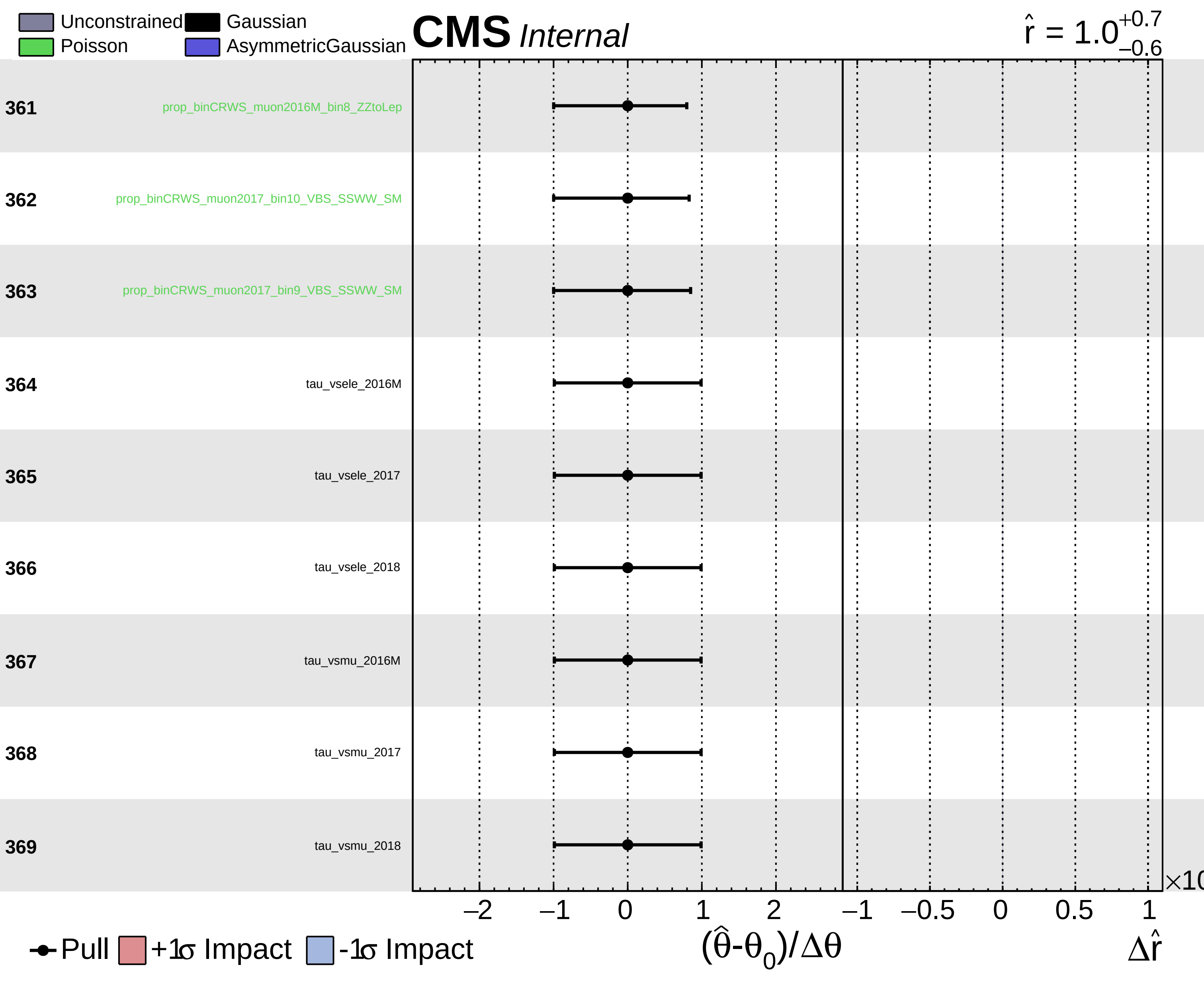

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.7}_{-0.6}$
271 prop_binCRTT_electron2017_bin4
272 jer_2017
273 prop_binCRTT_electron2016M_bin5
274 prop_binCRTT_electron2018_bin3
275 prop_binCRTT_electron2016M_bin4
276 prop_binCRTT_muon2017_bin3
277 prop_binSR_muon2018_bin0
278 prop_binCRTT_muon2016M_bin9
279 prop_binCRTT_electron2017_bin9
280 prop_binSR_electron2016M_bin0
281 prop_binCRTT_electron2017_bin5
282 prop_binCRWS_electron2016M_bin7_Fake
283 prop_binCRTT_electron2017_bin2
284 QCDScale_ZZtoLep
285 prop_binCRTT_muon2016M_bin5
286 prop_binCRWS_electron2017_bin9_WrongSign
287 puID
288 prop_binCRTT_electron2018_bin5
289 prop_binCRWS_muon2016M_bin8_Fake
290 prop_binCRTT_electron2018_bin0
291 prop_binSR_electron2016M_bin2_WZ
292 prop_binCRTT_muon2016M_bin11
293 prop_binCRWS_muon2017_bin9_Fake
294 prop_binSR_electron2016M_bin3_VG
295 prop_binSR_electron2016M_bin2_WrongSign
296 prop_binCRWS_electron2016M_bin8_Fake
297 lep_2016M
298 prop_binSR_electron2016M_bin2_TTTo2L2Nu
299 prop_binSR_electron2016M_bin3_TTTo2L2Nu
300 prop_binCRWS_electron2016M_bin9_Fake
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.7}_{-0.6}$
301 lep_2018
302 pdf_Tot_VG
303 prop_binCRWS_muon2016M_bin8_TTTo2L2Nu
304 prop_binCRWS_electron2016M_bin7_TTTo2L2Nu
305 prop_binCRWS_muon2016M_bin8_WrongSign
306 prop_binCRWS_muon2017_bin9_WrongSign
307 prop_binSR_electron2016M_bin3_ZZtoLep
308 prop_binCRWS_electron2016M_bin9_TTTo2L2Nu
309 prop_binCRWS_electron2016M_bin8_WrongSign
310 lep_2017
311 prop_binSR_electron2016M_bin2_VG
312 prop_binCRWS_electron2016M_bin9_WrongSign
313 prop_binCRWS_muon2017_bin9_TTTo2L2Nu
314 prop_binSR_electron2016M_bin2_ZZtoLep
315 prop_binSR_electron2016M_bin3_WrongSign
316 pdf_Tot_Triboson
317 prop_binCRWS_electron2017_bin9_TTTo2L2Nu
318 prop_binSR_electron2016M_bin3_TVX
319 prop_binCRWS_muon2017_bin10_TTTo2L2Nu
320 prop_binCRWS_muon2017_bin10_WrongSign
321 prop_binCRWS_electron2016M_bin7_WrongSign
322 prop_binCRWS_electron2016M_bin8_TTTo2L2Nu
323 prop_binSR_electron2016M_bin3_WpWpJJ_QCD
324 prop_binSR_electron2016M_bin3_VBS_SSWW_SM
325 prop_binSR_electron2016M_bin2_WpWpJJ_QCD
326 prop_binSR_electron2016M_bin2_TVX
327 prop_binCRWS_muon2017_bin10_VG
328 prop_binSR_electron2016M_bin3_Triboson
329 prop_binCRWS_electron2016M_bin7_TVX
330 prop_binSR_electron2016M_bin2_VBS_SSWW_SM
−2 −1 0 1 2
−0.2 −0.1 0 0.1 0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.7}_{-0.6}$
331 prop_binSR_electron2016M_bin2_Triboson
332 prop_binCRWS_electron2016M_bin8_VG
333 prop_binCRWS_muon2017_bin9_TVX
334 prop_binCRWS_electron2016M_bin8_TVX
335 prop_binCRWS_muon2016M_bin8_TVX
336 prop_binCRWS_muon2016M_bin8_VG
337 prop_binCRWS_electron2016M_bin9_Triboson
338 prop_binCRWS_electron2017_bin9_ZZtoLep
339 prop_binCRWS_electron2016M_bin8_ZZtoLep
340 prop_binCRWS_electron2016M_bin8_Triboson
341 prop_binCRWS_muon2017_bin9_ZZtoLep
342 prop_binCRWS_electron2016M_bin7_ZZtoLep
343 prop_binCRWS_electron2016M_bin9_TVX
344 prop_binCRWS_muon2017_bin9_Triboson
345 prop_binCRWS_electron2016M_bin7_Triboson
346 prop_binCRWS_electron2017_bin9_Triboson
347 prop_binCRWS_muon2016M_bin8_Triboson
348 prop_binCRWS_muon2017_bin10_TVX
349 prop_binCRWS_muon2017_bin10_Triboson
350 FES
351 prop_binCRWS_electron2016M_bin9_ZZtoLep
352 prop_binCRWS_electron2017_bin9_TVX
353 prop_binCRWS_muon2017_bin10_ZZtoLep
354 QCDScale_WZ
355 pdf_Tot_WZ
356 prop_binCRWS_electron2016M_bin7_VBS_SSWW_SM
357 prop_binCRWS_electron2016M_bin8_VBS_SSWW_SM
358 prop_binCRWS_electron2016M_bin9_VBS_SSWW_SM
359 prop_binCRWS_electron2017_bin9_VBS_SSWW_SM
360 prop_binCRWS_muon2016M_bin8_VBS_SSWW_SM
−2 −1 0 1 2
−10 −5 0 5 10
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.7}_{-0.6}$
361 prop_binCRWS_muon2016M_bin8_ZZtoLep
362 prop_binCRWS_muon2017_bin10_VBS_SSWW_SM
363 prop_binCRWS_muon2017_bin9_VBS_SSWW_SM
364 tau_vsele_2016M
365 tau_vsele_2017
366 tau_vsele_2018
367 tau_vsmu_2016M
368 tau_vsmu_2017
369 tau_vsmu_2018
−2 −1 0 1 2
−1 −0.5 0 0.5 1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$